

Biology and Environment

Chapter F9.4 — Completing the Ten 2023 Science Items: the Eyeball, the Banyan and Ocean Acidification

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NOTE · THIS CHAPTER CLOSES THE 2023 SCIENCE LIST

The last three of the ten 2023 General Science items are here: the **layers of the eyeball** (§2.4), **banyan prop roots** (§4.2) and **ocean acidification** (§5.3).

Unlike physics, biology is **not recall** for a Mechanical Engineer — it is closer to genuine new learning, and it is fact-dense rather than formula-based. So this chapter is built as **tables of facts organised by system**, with the three PYQ-evidenced sections given the fullest treatment.

Note also what the 2023 selection reveals: two of the three biology items were **plant morphology** and **environmental chemistry**, not human physiology. Candidates who study only the human body are aiming at the wrong third of the subject.

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0. How This Chapter Works

STUDY SESSION · CHAPTER F9.4 — FIVE SESSIONS

1. **Session 1** — The cell and cell division (§1) ★★★★★
2. **Session 2** — Human body systems, including **the eye** (§2) ★★★★★ **2023**
3. **Session 3** — Nutrition, vitamins and disease (§3) ★★★★★
4. **Session 4** — Plants, including **modified roots** (§4) ★★★★★ **2023**
5. **Session 5** — Ecology and environment, including **ocean acidification** (§5) ★★★★★ **2023**

Then 30 graded questions in §6. Budget **two and a half days**.

1. Session 1 — The Cell ★★★★★

The **cell** is the structural and functional unit of life. It was discovered by **Robert Hooke** in 1665, observing cork.

Prokaryotic cells have no true nucleus and no membrane-bound organelles — bacteria. **Eukaryotic** cells have both — plants, animals, fungi.

1.1 The organelles and their functions

Organelle	Function	Nickname
Nucleus	Controls the cell; contains DNA	—
Mitochondrion	Releases energy as ATP	Powerhouse of the cell
Ribosome	Protein synthesis	—
Chloroplast	Photosynthesis — plants only	—
Lysosome	Contains digestive enzymes	Suicidal bag
Golgi apparatus	Packaging and secretion	—
Endoplasmic reticulum	Transport within the cell	—
Vacuole	Storage; large and central in plant cells	—

MUST MASTER · PLANT VERSUS ANIMAL CELL, AND THE TWO DIVISIONS

A plant cell HAS a cell wall of cellulose, chloroplasts, and a large central vacuole. An animal cell has none of these but does have a centrosome.

The reliable wrong option reverses the cell wall — "a plant cell lacks a cell wall" is false, and it is offered because the *animal* cell is the one that lacks it.

	Mitosis	Meiosis
Occurs in	Somatic (body) cells	Reproductive cells
Daughter cells	Two , genetically identical	Four
Chromosome number	Unchanged	Halved — reduction division
Purpose	Growth and repair	Formation of gametes

A normal human somatic cell has **46 chromosomes — 23 pairs**. Sex chromosomes: **XX female, XY male**. DNA's double-helix structure was described by **Watson and Crick**.

CHECK YOURSELF · SESSION 1

1. Who discovered the cell, and in what material?
2. Which organelle is the powerhouse? Which synthesises protein? Which is the suicidal bag?
3. Name three things a plant cell has that an animal cell does not
4. Give four differences between mitosis and meiosis
5. How many chromosomes in a human somatic cell?

2. Session 2 — Human Body Systems ★★★★★

2.1 Digestion and circulation

Organ	Function
Mouth	Salivary amylase begins starch digestion
Stomach	HCl and pepsin — protein digestion
Small intestine	Main site of absorption
Large intestine	Absorption of water
Liver	The largest gland ; secretes bile
Pancreas	Both exocrine (enzymes) and endocrine (insulin)

Circulation. The heart has **four chambers** — two atria and two ventricles.

MUST MASTER · ARTERIES, VEINS AND THE TWO EXCEPTIONS

Arteries carry blood AWAY from the heart; veins carry blood TO the heart. That is the definition — direction, not oxygen content.

Oxygen content follows as a general rule with **two exceptions**:

- The **pulmonary artery** carries **deoxygenated** blood, from the heart to the lungs
- The **pulmonary vein** carries **oxygenated** blood, from the lungs to the heart

So a question asking which artery carries deoxygenated blood is answered by the **pulmonary artery**, and a statement that "all arteries carry oxygenated blood" is wrong.

Blood: RBC contain **haemoglobin** and live about 120 days; **WBC** provide immunity; **platelets** enable clotting. **O negative is the universal donor; AB positive the universal recipient.**

2.2 Excretion, nerves and hormones

- **Kidney** — the functional unit is the **nephron**; the nitrogenous waste is **urea**
- **Brain** — the **cerebrum** is the largest part and governs thinking; the **cerebellum** governs

balance and coordination; the **medulla oblongata** controls involuntary functions such as heartbeat and breathing. **Reflex action** is mediated by the **spinal cord**, not the brain

Gland	Hormone	Deficiency or excess
Pituitary	Growth hormone	The master gland
Thyroid	Thyroxine — requires iodine	Iodine deficiency causes goitre
Pancreas	Insulin	Deficiency causes diabetes mellitus
Adrenal	Adrenaline	The emergency hormone

An adult human has **206 bones**. The **femur** is the largest; the **stapes**, in the ear, the smallest.

2.3 The ear

Outer, middle and inner. The middle ear contains **three bones — malleus, incus and stapes**. In the inner ear the **cochlea** serves **hearing** and the **semicircular canals** serve **balance**.

2.4 The eye — the 2023 item

MUST MASTER · THE THREE LAYERS OF THE EYEBALL, OUTSIDE INWARDS

This is a 2023 question. The wall of the eyeball has **three coats**, and the order matters:

Order	Layer	Nature
1 — outermost	Sclera	Tough, white, protective. Its transparent front part is the cornea
2 — middle	Choroid	Vascular and pigmented; supplies blood and prevents internal reflection
3 — innermost	Retina	Light-sensitive , carrying the rods and cones

Mnemonic for the order: **S-C-R**, outside in — **Sclera**, **Choroid**, **Retina**. Alphabetical, which makes it unusually easy.

The trap is any option placing the **choroid innermost** or the **retina outermost**.

Inside the retina: **rods** give vision in **dim light**, in black and white; **cones** give **colour** vision in bright light. The **blind spot** is where the optic nerve leaves and has neither.

Other parts: the **iris** controls the size of the **pupil** and hence the light admitted; the **lens** focuses; the aqueous and vitreous humours fill the chambers. Defects of vision and their corrections are in **F9.2 §1.5**.

CHECK YOURSELF · SESSION 2

1. Which is the largest gland? Which organ is the main site of absorption?
2. State the artery-vein rule and name the two exceptions
3. Universal donor and universal recipient blood groups?
4. Which part of the brain controls balance? Which controls heartbeat? What mediates reflex action?
5. Name the three layers of the eyeball from outside inwards
6. Rods and cones — which does what?

3. Session 3 — Nutrition and Disease ★★★★★

3.1 Vitamins and deficiency diseases

MUST MASTER · FAT-SOLUBLE VERSUS WATER-SOLUBLE, AND THE DEFICIENCY LIST

Fat-soluble vitamins: A, D, E and K. Everything else — the **B group and C** — is **water-soluble**. That four-letter list is asked directly.

Vitamin	Chemical name	Deficiency disease
A	Retinol	Night blindness , xerophthalmia
B1	Thiamine	Beri-beri
B3	Niacin	Pellagra
B12	Cobalamin	Pernicious anaemia
C	Ascorbic acid	Scurvy
D	Calciferol	Rickets in children, osteomalacia in adults
E	Tocopherol	Sterility
K	—	Defective blood clotting

Mineral deficiencies: iron » **anaemia** · iodine » **goitre** · calcium and vitamin D » rickets.
Protein deficiency causes **kwashiorkor** and **marasmus**.

3.2 Diseases by causative organism

Cause	Diseases
Bacteria	Tuberculosis , cholera, typhoid, tetanus, leprosy, plague, diphtheria
Virus	Influenza, AIDS , polio, measles, mumps, rabies, dengue , chikungunya, hepatitis
Protozoa	Malaria (<i>Plasmodium</i>), amoebic dysentery, kala-azar
Fungus	Ringworm, athlete's foot

MUST MASTER · THE MOSQUITO VECTORS

Mosquito	Transmits
Anopheles	Malaria
Aedes aegypti	Dengue, chikungunya, yellow fever, Zika
Culex	Filariasis

And the classification trap: **malaria is protozoan, not bacterial or viral**, while **dengue is viral, not protozoan**. Both are mosquito-borne, which is exactly why the causative organisms get swapped.

CHECK YOURSELF · SESSION 3

1. Which four vitamins are fat-soluble?
2. Name the deficiency diseases of vitamins A, B1, C and D
3. Iron deficiency causes what? Iodine deficiency?
4. Is malaria bacterial, viral or protozoan? And dengue?
5. Match Anopheles, Aedes and Culex to their diseases

4. Session 4 — Plants ★★★★★

4.1 Photosynthesis and transport

Photosynthesis occurs in the **chloroplast** using **chlorophyll** and sunlight: carbon dioxide and water become glucose and oxygen. It is **endothermic** — it absorbs energy, which is F9.3 §5.1.

Transport: xylem carries **water and minerals upwards**; **phloem** carries **food**, and it moves in **both directions**.

4.2 Modified roots — the 2023 item

MUST MASTER · THE FOUR MODIFIED ROOTS, WITH THEIR PLANTS

This is a 2023 question, and the four are constantly interchanged.

Modification	Purpose	Plant
Prop roots	Aerial roots growing downwards from branches to support heavy spreading branches	BANYAN
Stilt roots	Growing obliquely from the lower stem into the soil for support	Maize, sugarcane, screwpine
Pneumatophores (respiratory roots)	Growing upwards out of waterlogged soil for gas exchange	MANGROVES
Storage roots	Storing food	Carrot, sweet potato, radish

The distinction that decides the question is **direction**:

- **Prop roots grow DOWN, from the branches** — banyan
- **Stilt roots grow DOWN and outward, from the lower stem** — sugarcane
- **Pneumatophores grow UP, out of the mud** — mangrove

So "aerial roots hanging from the branches of a large tree" is unambiguously the **banyan**, and "roots growing vertically upward from waterlogged soil" is the **mangrove**.

Modified stems, for contrast, since they are offered as distractors: **potato** is a *tuber*, **ginger** a *rhizome*, **onion** a *bulb*.

Nitrogen fixation: leguminous plants carry **root nodules** containing *Rhizobium*.

4.3 Plant hormones and tropisms

Hormone	Effect
Auxin	Cell elongation; responsible for phototropism
Gibberellin	Stem elongation
Cytokinin	Cell division
Abscisic acid	Stress response, closing stomata; growth inhibitor
Ethylene	Ripening of fruit

Tropisms: photo (light), geo (gravity), hydro (water), thigmo (touch), chemo (chemicals).

CHECK YOURSELF · SESSION 4

1. Where does photosynthesis occur, and is it exothermic or endothermic?
2. Xylem and phloem — what does each carry, and in which direction?
3. Name the plant for prop roots, stilt roots, pneumatophores and storage roots
4. Which direction do prop roots grow? Which way do pneumatophores grow?
5. Which hormone ripens fruit? Which causes phototropism?

5. Session 5 — Ecology and Environment ★★★★★

5.1 Ecosystem structure

An **ecosystem** has **biotic** components — **producers, consumers** and **decomposers** — and **abiotic** components.

MUST MASTER · THE TEN PER CENT LAW AND UNIDIRECTIONAL FLOW

Only about 10% of the energy at one trophic level passes to the next — Lindeman's ten per cent law. The rest is lost as heat and in metabolism.

Two consequences that are examined:

1. **Energy flow through an ecosystem is UNIDIRECTIONAL**, not cyclic. Nutrients cycle; **energy does not**. An option saying energy flow is bidirectional or cyclic is wrong
2. Food chains are therefore **short** — usually four or five links — because so little energy survives

Biomagnification is the increasing concentration of **non-biodegradable** substances such as DDT and mercury at successively higher trophic levels. The word to check in the option is **non-biodegradable** — biodegradable substances do not biomagnify.

5.2 Pollution

Type	Key indicators and facts
Air	PM2.5 and PM10, sulphur dioxide, oxides of nitrogen, carbon monoxide
Water	BOD — biochemical oxygen demand. Higher BOD means more organic pollution
Eutrophication	Nutrient enrichment by nitrates and phosphates » algal bloom » oxygen depletion » fish die

Note the link back to F9.3 §4.1: **phosphates from detergents** are a classic contributor to eutrophication, which is part of why detergent biodegradability matters.

Acid rain comes from oxides of **sulphur and nitrogen** forming sulphuric and nitric acids.

5.3 Ocean acidification — the 2023 item

MUST MASTER · WHAT ACIDIFICATION MEANS, AND WHAT IT DOES NOT

The ocean absorbs a large share of the carbon dioxide humans emit. Dissolved **carbon dioxide forms carbonic acid**, which releases hydrogen ions and **lowers the pH of seawater**.

Consequence: less carbonate is available for shell and skeleton formation, so the organisms harmed are the **calcifying** ones — **corals**, molluscs and shelled plankton. Coral bleaching and reef weakening follow.

And now the point the exam is most likely to test. Ocean pH has fallen from roughly **8.2 to about 8.1** since pre-industrial times. So:

Seawater is still ALKALINE. "Acidification" means moving TOWARDS acidic, not having become acidic.

Any statement that the ocean has become acidic, or that its pH is below 7, is **wrong** — and it is the intuitive reading of the word. The cause is **carbon dioxide**, not sulphur dioxide, which is the agent of acid *rain*.

5.4 Climate, ozone and the conventions

Greenhouse gases: carbon dioxide, **methane**, nitrous oxide, water vapour, CFCs. Methane has a higher global warming potential per molecule than carbon dioxide.

Ozone. Stratospheric ozone absorbs **ultraviolet** radiation and is beneficial; tropospheric ozone is a **pollutant**. Depletion is caused by **CFCs**, and the ozone hole appears over **Antarctica**.

Agreement	Year	Subject
Ramsar Convention	1971	Wetlands
CITES	1973	Trade in endangered species
Montreal Protocol	1987	Ozone-depleting substances
UNFCCC and CBD	1992	Climate change; biological diversity
Kyoto Protocol	1997	Emission reduction commitments
Paris Agreement	2015	Climate change

COMMON UPSC TRAP · MONTREAL VERSUS KYOTO VERSUS PARIS

Three protocols, three different subjects, routinely swapped:

- **Montreal 1987 » OZONE**
- **Kyoto 1997 » climate, binding emission targets**
- **Paris 2015 » climate, nationally determined contributions**

Montreal is the ozone one. If the stem mentions the ozone layer or CFCs and the options include Kyoto or Paris, they are there to catch you. And **Ramsar is wetlands**, not forests.

India: Project Tiger began in **1973**, **Project Elephant** in 1992. Protected areas comprise national parks, wildlife sanctuaries and biosphere reserves.

CHECK YOURSELF · SESSION 5

1. State the ten per cent law. Is energy flow cyclic or unidirectional?
2. What kind of substances biomagnify?
3. What does a high BOD indicate?
4. Explain eutrophication in sequence, and name two nutrients responsible
5. What causes ocean acidification, which organisms suffer, and is seawater now acidic?
6. Match Ramsar, Montreal, Kyoto and Paris to their subjects

6. Graded Practice — 30 Questions ★★★★★

6.1 Level 1 — Recognition

- Q1.** The structural and functional unit of life is the (a) tissue (b) organ (c) organelle (d) cell
- Q2.** The powerhouse of the cell is the (a) ribosome (b) mitochondrion (c) nucleus (d) lysosome
- Q3.** The outermost layer of the human eyeball is the (a) retina (b) choroid (c) sclera (d) vitreous humour
- Q4.** Protein synthesis within a cell takes place in the (a) ribosome (b) lysosome (c) vacuole (d) Golgi apparatus
- Q5.** The number of chromosomes in a normal human somatic cell is (a) 23 (b) 48 (c) 44 (d) 46
- Q6.** Deficiency of vitamin C causes (a) rickets (b) scurvy (c) beri-beri (d) night blindness
- Q7.** Malaria is caused by a (a) bacterium (b) virus (c) protozoan (d) fungus
- Q8.** Prop roots are characteristically seen in the (a) banyan (b) maize (c) carrot (d) mangrove
- Q9.** The functional unit of the kidney is the (a) neuron (b) alveolus (c) villus (d) nephron
- Q10.** The universal donor blood group is (a) AB positive (b) O negative (c) A positive (d) B negative

6.2 Level 2 — Understanding

- Q11.** The three layers of the human eyeball, from outside inwards, are (a) retina, choroid, sclera (b) choroid, sclera, retina (c) sclera, choroid, retina (d) cornea, retina, choroid
- Q12.** Rods in the retina are responsible for (a) vision in dim light (b) colour vision (c) focusing the image (d) controlling the size of the pupil
- Q13.** Which part of the brain controls balance and coordination? (a) Cerebrum (b) Medulla oblongata (c) Hypothalamus (d) Cerebellum
- Q14.** The master gland of the human body is the (a) thyroid (b) pituitary (c) adrenal (d) pancreas
- Q15.** Deficiency of iodine causes (a) diabetes (b) anaemia (c) goitre (d) rickets
- Q16.** Pneumatophores, or respiratory roots, are found in (a) mangroves (b) the banyan (c) sugarcane (d) the money plant
- Q17.** According to the ten per cent law, the proportion of energy transferred to the next trophic level is about (a) 90% (b) 50% (c) 1% (d) 10%
- Q18.** Ocean acidification is caused mainly by the absorption of (a) sulphur dioxide (b) carbon dioxide (c) methane (d) nitrogen
- Q19.** Eutrophication of a water body results in (a) an increase in dissolved oxygen (b) reduced algal growth (c) an algal bloom followed by oxygen depletion (d) a rise in pH to 12

Q20. A high biochemical oxygen demand in a sample of water indicates (a) a high level of organic pollution (b) unusually clean water (c) high dissolved oxygen (d) low bacterial activity

6.3 Level 3 — Recruitment Test standard

Q21. Which one of the following statements is **not** correct? (a) Arteries generally carry oxygenated blood, the pulmonary artery being an exception (b) Rods are responsible for vision in dim light and cones for colour vision (c) The retina is the innermost layer of the eyeball (d) The choroid is the innermost layer of the eyeball

Q22. Consider the following statements about ocean acidification:

1. It is caused by the dissolution of atmospheric carbon dioxide, forming carbonic acid.
2. It harms calcifying organisms such as corals and molluscs.
3. Seawater has already become acidic, its pH having fallen below 7.

Which of the above statements are correct? (a) 1 and 3 only (b) 1 and 2 only (c) 2 and 3 only (d) 1, 2 and 3

Q23. Match List-I with List-II and select the correct answer using the code given below:

List-I	List-II
A. Prop roots	1. Banyan
B. Stilt roots	2. Sugarcane
C. Pneumatophores	3. Mangrove
D. Storage root	4. Carrot

(a) A-4 B-3 C-2 D-1 (b) A-1 B-3 C-2 D-4 (c) A-1 B-2 C-3 D-4 (d) A-2 B-1 C-4 D-3

Q24. Which of the following vitamins are fat-soluble? (a) A, D, E and K (b) B and C only (c) C and D only (d) All vitamins are fat-soluble

Q25. Consider the following statements:

1. Mitosis produces two genetically identical daughter cells.
2. Meiosis halves the chromosome number and produces gametes.
3. A plant cell lacks a cell wall but possesses chloroplasts.

Which of the above statements are correct? (a) 1 and 3 only (b) 2 and 3 only (c) 1, 2 and 3 (d) 1 and 2 only

Q26. Dengue is transmitted by the (a) Anopheles mosquito (b) Aedes aegypti mosquito (c) Culex mosquito (d) common housefly

Q27. The Montreal Protocol relates to (a) climate change (b) biological diversity (c) substances that deplete the ozone layer (d) wetlands

Q28. Which one of the following statements is **not** correct? (a) Energy flow in an ecosystem is bidirectional (b) Biomagnification concerns non-biodegradable substances (c) Decomposers return nutrients to the soil (d) Only about ten per cent of the energy at one trophic level passes to the next

Q29. The largest gland in the human body is the (a) pancreas (b) thyroid (c) pituitary (d) liver

Q30. Insulin is secreted by the (a) liver (b) pancreas (c) adrenal gland (d) thyroid gland

7. Answers and Explanations

7.1 Level 1

Q	Ans	Explanation
1	d	The cell , discovered by Robert Hooke in 1665
2	b	The mitochondrion , which releases energy as ATP
3	c	The sclera . Remember S-C-R outside in — sclera, choroid, retina. The cornea is the transparent front of the sclera
4	a	The ribosome
5	d	46 , that is 23 pairs. Option (a) gives the number of pairs, not chromosomes
6	b	Scurvy . Rickets is vitamin D, beri-beri is B1, night blindness is A
7	c	A protozoan , <i>Plasmodium</i> . Dengue , by contrast, is viral — the two are swapped because both are mosquito-borne
8	a	The banyan . Sugarcane has stilt roots, mangroves pneumatophores, carrot a storage root
9	d	The nephron . The neuron belongs to the nervous system, the alveolus to the lungs, the villus to the intestine
10	b	O negative is the universal donor; AB positive the universal recipient

7.2 Level 2

Q	Ans	Explanation
11	c	Sclera, choroid, retina — S-C-R, and conveniently alphabetical outside in
12	a	Dim light , in black and white. Cones handle colour in bright light
13	d	The cerebellum . The cerebrum governs thinking; the medulla controls heartbeat and breathing
14	b	The pituitary
15	c	Goitre , because iodine is required to make thyroxine
16	a	Mangroves — these roots grow upwards out of waterlogged soil for gas exchange
17	d	About 10% , Lindeman's law. It is why food chains are short
18	b	Carbon dioxide , which forms carbonic acid. Sulphur dioxide causes acid <i>rain</i> — a different phenomenon
19	c	Algal bloom followed by oxygen depletion , which kills fish. Note that dissolved oxygen <i>falls</i> , so (a) is the reverse
20	a	High organic pollution — more organic matter means more oxygen consumed by decomposers

7.3 Level 3

Q	Ans	Explanation
21	d	The choroid is the MIDDLE layer , not the innermost. The retina is innermost, which (c) correctly states — so (d) is the contradiction
22	b	1 and 2 only. Statement 3 is false and is the central trap: ocean pH has fallen from about 8.2 to 8.1, so seawater remains alkaline . "Acidification" means moving <i>towards</i> acidic
23	c	A-1 B-2 C-3 D-4 — banyan, sugarcane, mangrove, carrot. Direction is the discriminator: prop roots down from branches, pneumatophores up from mud
24	a	A, D, E and K. The B group and C are water-soluble
25	d	1 and 2 only. Statement 3 is false — a plant cell has a cell wall. It is the <i>animal</i> cell that lacks one
26	b	<i>Aedes aegypti</i> , which also carries chikungunya, yellow fever and Zika. <i>Anopheles</i> carries malaria
27	c	Ozone-depleting substances. Kyoto and Paris are climate; Ramsar is wetlands
28	a	Energy flow is unidirectional , not bidirectional. Nutrients cycle; energy does not. The other three are correct
29	d	The liver
30	b	The pancreas

THINK LIKE UPSC · THE FOUR SHAPES IN THIS CHAPTER

- The ordered list.** Q11 and Q21 both test the sclera–choroid–retina order. Any three-layer structure invites a scrambled-order question, so learn the sequence with its direction
- The word taken literally.** Q22. "Acidification" sounds like "has become acidic". It means *moving towards* acidic. Where a term implies a state, check whether it actually describes a trend
- The vector-versus-organism swap.** Q7 and Q26. Malaria and dengue share mosquito transmission but differ completely in causative organism — protozoan versus virus
- The reversed possession.** Q25's statement 3 and Q28's option (a). The examiner takes a true fact about one thing and attaches it to its opposite. Cell wall belongs to the *plant*; unidirectional flow to *energy*

8. One-Minute Revision

ONE-MINUTE REVISION · CHAPTER F9.4

CELL unit of life · **Robert Hooke 1665, cork** · prokaryote has no true nucleus · **mitochondrion = powerhouse** · **ribosome = protein synthesis** · **lysosome = suicidal bag** · chloroplast = photosynthesis · **plant cell HAS cell wall, chloroplast, large vacuole**

DIVISION **mitosis** = somatic, **2 identical**, number unchanged · **meiosis** = gametes, **4 cells**, **number halved** · human somatic cell **46 chromosomes**, XX female, XY male

DIGESTION salivary amylase · stomach HCl and pepsin · **small intestine = main absorption** · large intestine absorbs **water** · **liver = largest gland**, secretes bile · pancreas secretes **insulin**

CIRCULATION **arteries carry blood AWAY**, **veins TOWARDS the heart** · exceptions **pulmonary artery carries deoxygenated**, **pulmonary vein oxygenated** · **O negative universal donor**, **AB positive universal recipient** · heart has **4 chambers**

BRAIN cerebrum = thinking, largest · **cerebellum = balance and coordination** · **medulla = heartbeat and breathing** · **reflex action via the SPINAL CORD**

GLANDS **pituitary = master** · thyroid = thyroxine, needs **iodine**, deficiency **goitre** · pancreas = **insulin**, deficiency **diabetes** · adrenal = adrenaline, emergency hormone

EYE — THREE LAYERS, OUTSIDE IN: SCLERA » CHOROID » RETINA (S-C-R) · cornea = transparent front of sclera · **rods = dim light**, **cones = colour** · iris controls the pupil · blind spot has neither rods nor cones

EAR middle ear bones **malleus, incus, stapes** · **cochlea hearing**, **semicircular canals balance** · **stapes = smallest bone**, femur largest · **206 bones**

FAT-SOLUBLE VITAMINS A, D, E, K — the rest are water-soluble · **A night blindness** · **B1 beri-beri** · **B3 pellagra** · **C scurvy** · **D rickets** · **K clotting** · **iron » anaemia** · **iodine » goitre** · protein » kwashiorkor, marasmus

DISEASES bacterial: TB, cholera, typhoid, tetanus, leprosy · viral: **dengue**, AIDS, polio, rabies · protozoan: **malaria**, kala-azar · **Anopheles » malaria** · **Aedes » dengue, chikungunya, Zika** · **Culex » filariasis**

PLANTS photosynthesis in the **chloroplast**, **endothermic** · **xylem water UP** · **phloem food BOTH ways** · **Rhizobium** in root nodules

MODIFIED ROOTS **prop » BANYAN**, down from branches · **stilt » sugarcane, maize**, from lower stem · **pneumatophores » MANGROVE**, growing **UP** out of mud · **storage » carrot** · **MODIFIED STEMS** potato tuber · ginger rhizome · onion bulb

HORMONES auxin » phototropism · **ethylene** » ripening · abscisic acid » stress and inhibition

ECOLOGY 10% law · energy flow **UNIDIRECTIONAL**, nutrients cycle · **biomagnification of NON-biodegradable** substances, DDT and mercury

POLLUTION high BOD = high organic pollution · **eutrophication**: nitrates and phosphates » algal bloom » **oxygen depletion** » fish die · acid rain from **sulphur and nitrogen** oxides

OCEAN ACIDIFICATION dissolved **CO₂** » carbonic acid » pH falls from about **8.2 to 8.1** · **seawater is STILL ALKALINE** — acidification means moving **towards** acidic · harms **calcifying** organisms: corals, molluscs, shelled plankton

OZONE stratospheric ozone absorbs **UV** and is good; tropospheric is a pollutant · depleted by **CFCs** · hole over **Antarctica**

AGREEMENTS Ramsar **1971 wetlands** · CITES 1973 species trade · **Montreal 1987 OZONE** · UNFCCC and CBD 1992 · Kyoto 1997 · **Paris 2015 climate** · **Project Tiger 1973**

9. Five-Minute Wall Sheet

FIVE-MINUTE WALL SHEET · CHAPTER F9.4

EYE: S-C-R OUTSIDE IN — SCLERA, CHOROID, RETINA · rods dim, cones colour

ARTERIES AWAY, VEINS TOWARDS · pulmonary artery is the deoxygenated exception

O NEGATIVE DONOR · AB POSITIVE RECIPIENT

CEREBELLUM = BALANCE · MEDULLA = HEARTBEAT · REFLEX = SPINAL CORD

PITUITARY = MASTER · IODINE » GOITRE · INSULIN FROM PANCREAS · LIVER LARGEST GLAND

A D E K ARE FAT-SOLUBLE

MALARIA IS PROTOZOAN · DENGUE IS VIRAL · Anopheles malaria, Aedes dengue, Culex filariasis

PROP = BANYAN (down from branches) · PNEUMATOPHORE = MANGROVE (up from mud) · STILT = SUGARCANE

MITOSIS 2 IDENTICAL · MEIOSIS 4 HALVED · PLANT CELL HAS A CELL WALL

10% LAW · ENERGY UNIDIRECTIONAL, NUTRIENTS CYCLE

HIGH BOD = DIRTY WATER · EUTROPHICATION KILLS BY OXYGEN DEPLETION

OCEAN ACIDIFICATION = CO₂ » CARBONIC ACID » pH 8.2 TO 8.1 · STILL ALKALINE · HARMS CALCIFIERS

MONTREAL 1987 = OZONE · KYOTO 1997 & PARIS 2015 = CLIMATE · RAMSAR 1971 = WETLANDS

ONE LINE FOR THE INTERVIEW Ocean acidification is a good example of a term that misleads by sounding absolute: seawater is still alkaline at about pH 8.1, but the shift of roughly 0.1 units since pre-industrial times represents about a thirty per cent rise in hydrogen ion concentration, which is already enough to impair shell formation in calcifying organisms.

ACTION · THE SCIENCE HALF OF MODULE 9 IS COMPLETE

All **ten** of the 2023 General Science items are now covered across F9.1 to F9.4.

Next: F9.5 to F9.7 on Computer Applications — and these switch back to a genuine **Foundation layer**, not a recall layer. Blueprint §4.4 is explicit that number-system conversions are easy but the C, HTML and operating-system items *"need actual targeted preparation. Budget real hours there, not zero."*

Before you move on, state without looking: the three layers of the eyeball in order; which mosquito carries dengue; where prop roots grow from and in which plant; what causes ocean acidification and whether seawater is now acidic; and which protocol concerns ozone.